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Motor grader and method of controlling motor grader

Abstract

A motor grader includes an operation apparatus, a swing circle, a blade supported on the swing circle, a first actuator arranged along a longitudinal direction of the blade, the first actuator moving the blade in a lateral direction with respect to the swing circle, and a controller that causes the first actuator to operate. The controller receives an operation signal from the operation apparatus and causes the first actuator to operate based on the received operation signal such that a position of the blade with respect to the swing circle comes closer to a neutral position of the blade with respect to the swing circle.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a motor grader and a method of controlling a motor grader.

BACKGROUND ART

(2) A motor grader has conventionally been known as a work vehicle. The motor grader includes a work implement including a draw bar, a swing circle, a blade, and the like.

(3) For example, U.S. Patent Publication No. 2018/0106014 (PTL 1) discloses a motor grader that automatically moves a work implement to a forwarding position when an operator operates a switch for automatic transport control provided in a control lever. According to such a

configuration, PTL 1 saves time and trouble of an operator in forwarding a motor grader with transport means.
CITATION LIST
Patent Literature
(4) PTL 1: U.S. Patent Publication No. 2018/0106014
SUMMARY OF INVENTION
Technical Problem
(5) The operator moves a blade toward or away from a front frame by causing a pair of left and right lift cylinders to operate during works. At that time, when the blade is not located at a neutral position with respect to a swing circle, an amount of upward/downward movement of a left end of the blade at the time when the operator operates a left lift cylinder by a certain amount is different from an amount of upward/downward movement of a right end of the blade at the time when the operator operates a right lift cylinder by an amount equal to the amount of movement of the left lift cylinder.
(6) Therefore, as the blade is located at a position more distant from the neutral position, it is more difficult for an operator who is not skilled in operation to move the blade to a desired position.
(7) The present disclosure was made in view of the problem above, and an object thereof is to provide a motor grader capable of lessening burdens imposed on an operator in operating a work implement and a method of controlling a motor grader.
Solution to Problem
(8) According to one aspect of the present disclosure, a motor grader includes an operation apparatus, a swing circle, a blade supported on the swing circle, a first actuator arranged along a longitudinal direction of the blade, the first actuator moving the blade in a lateral direction with respect to the swing circle, and a controller that causes the first actuator to operate. The controller receives an operation signal from the operation apparatus and causes the first actuator to operate based on the received operation signal such that a position of the blade with respect to the swing circle comes closer to a neutral position of the blade with respect to the swing circle.
(9) According to another aspect of the present disclosure, a method of controlling a motor grader is provided. The motor grader includes an operation apparatus, a swing circle, a blade supported on the swing circle, and a first actuator arranged along a longitudinal direction of the blade, the first actuator moving the blade in a lateral direction with respect to the swing circle. The method includes receiving an operation signal from the operation apparatus based on an operation performed onto the operation apparatus and causing the first actuator to operate based on reception of the operation signal from the operation apparatus such that a position of the blade with respect to the swing circle comes closer to a neutral position of the blade with respect to the swing circle.
Advantageous Effects of Invention
(10) According to the present disclosure, burdens imposed on an operator in operating a work implement can be lessened.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a perspective view schematically showing a construction of a motor grader.
(2) FIG. **2** is an enlarged perspective view showing a main part of a work implement of the motor grader.
(3) FIG. **3** is a functional block diagram illustrating a functional configuration of a control system of the motor grader.
(4) FIG. **4** is a schematic diagram showing a state in which each of a blade and a draw bar is located at a neutral position.

(5) FIG. **5** is a diagram for illustrating an operation for causing the blade to make transition to the neutral position.
(6) FIG. **6** is a diagram for illustrating an operation for causing the draw bar to make transition to the neutral position.
(7) FIG. **7** is a diagram for illustrating an advantage obtained by causing the work implement to make transition to the neutral position during works.
(8) FIG. **8** is a flowchart for illustrating a flow of processing performed in the motor grader.
(9) FIG. **9** is a flowchart for illustrating details of processing in step S**4** in FIG. **8**.
DESCRIPTION OF EMBODIMENTS
(10) A motor grader work vehicle according to an embodiment of the present invention will be described below with reference to the drawings. In the description below, the same elements have the same reference characters allotted and their labels and functions are also the same. Therefore, detailed description thereof will not be repeated.
(11) <A. Schematic Construction of Motor Grader>
(12) FIG. **1** is a perspective view schematically showing a construction of a motor grader **1** based on the present embodiment. As shown in FIG. **1**, motor grader **1** mainly includes a front wheel **11**, a rear wheel **12**, a vehicular body frame **2**, a cab **3**, and a work implement **4**. Motor grader **1** includes components such as an engine arranged in an engine compartment **6**. Work implement **4** includes a blade **42**. Motor grader **1** does such works as land-grading works, snow removal works, light cutting, and mixing of materials with blade **42**.
(13) In the description of the drawings below, a direction in which motor grader **1** travels in straight lines is referred to as a fore/aft direction of motor grader **1**. In the fore/aft direction of motor grader **1**, a side where front wheel **11** is arranged with respect to work implement **4** is defined as the fore direction. In the fore/aft direction of motor grader **1**, a side where rear wheel **12** is arranged with respect to work implement **4** is defined as the aft direction.
(14) A lateral direction of motor grader **1** is a direction orthogonal to the fore/aft direction in a plan view. A right side and a left side in the lateral direction in facing front are defined as a right direction and a left direction, respectively. An upward/downward direction of motor grader **1** is a direction orthogonal to the plane defined by the fore/aft direction and the lateral direction. A side in the upward/downward direction where the ground is located is defined as a lower side and a side where the sky is located is defined as an upper side.
(15) The fore/aft direction refers to a fore/aft direction of an operator who sits at an operator's seat in cab **3**. The lateral direction refers to a lateral direction of the operator who sits at the operator's seat. The lateral direction refers to a direction of a vehicle width of motor grader **1**. The upward/downward direction refers to an upward/downward direction of the operator who sits at the operator's seat. A direction in which the operator sitting at the operator's seat faces is defined as the fore direction and a direction behind the operator sitting at the operator's seat is defined as the aft direction. A right side and a left side at the time when the operator sitting at the operator's seat faces front are defined as the right direction and the left direction, respectively. A foot side of the operator who sits at the operator's seat is defined as a lower side, and a head side is defined as an upper side.
(16) In the present example, the fore direction corresponds to a negative direction along an X axis in the drawing. The aft direction corresponds to a positive direction along the X axis. The left direction corresponds to the positive direction along a Y axis. The right direction corresponds to the negative direction along the Y axis. The upward direction corresponds to the positive direction along a Z axis. The downward direction corresponds to the negative direction along the Z axis.
(17) Vehicular body frame **2** extends in the fore/aft direction. Vehicular body frame **2** includes a rear frame **21** and a front frame **22**.
(18) Rear frame **21** supports an exterior cover **25** and components such as an engine arranged in engine compartment **6**. Exterior cover **25** covers engine compartment **6**. For example, each of four

rear wheels **12** is attached to rear frame **21** as being rotatably driven by driving force from the engine.
(19) Front frame **22** is attached in front of rear frame **21**. Front frame **22** is pivotably coupled to rear frame **21**. Front frame **22** extends in the fore/aft direction. Front frame **22** includes a base end portion coupled to rear frame **21** and a tip end portion opposite to the base end portion. The base end portion of front frame **22** is coupled to the tip end portion of rear frame **21** with a vertical central pin being interposed.
(20) An articulation cylinder (not shown) is attached between front frame **22** and rear frame **21**. Front frame **22** is provided as being pivotable with respect to rear frame **21** owing to extending and retracting of the articulation cylinder. The articulation cylinder is provided as being extensible and retractable in response to an operation of a control lever provided in cab **3**.
(21) For example, two front wheels **11** are rotatably attached to the tip end portion of front frame **22**. Front wheel **11** is attached to front frame **22** as being revolvable owing to extending and retracting of a steering cylinder (not shown). Motor grader **1** can change its direction of travel owing to extending and retracting of the steering cylinder. The steering cylinder can extend and retract in response to an operation of a steering wheel or a steering control lever provided in cab **3**.
(22) A counter weight **51** is attached to a front end of vehicular body frame **2**. Counter weight **51** represents one type of attachments to be attached to front frame **22**. Counter weight **51** is attached to front frame **22** in order to increase a downward load to be applied to front wheel **11** to allow steering and to increase a pressing load on blade **42**.
(23) Cab **3** is carried on front frame **22**. In cab **3**, an operation portion (not shown) such as a steering wheel, a gear shift lever, a lever for controlling work implement **4**, a brake, an accelerator pedal, an inching pedal, and various switches is provided. Cab **3** may be carried on rear frame **21**.
(24) <B. Construction of Main Part of Work Implement>
(25) FIG. **2** is an enlarged perspective view showing a main part of work implement **4** of motor grader **1** shown in FIG. **1**. As shown in FIG. **2**, work implement **4** mainly includes a draw bar **40**, a swing circle **41**, and blade **42**.
(26) Draw bar **40** is arranged below front frame **22**. Draw bar **40** is moved by a pair of lift cylinders **44** and **45** to move in a direction toward front frame **22** (a direction in which blade **42** moves away from the ground) and a direction away from front frame **22**.
(27) Draw bar **40** has a front end portion coupled to the tip end portion of front frame **22** with a ball bearing portion **402**. Draw bar **40** has the front end portion swingably attached to the tip end portion of front frame **22**. Draw bar **40** has a rear end portion supported on front frame **22** by lift cylinders **44** and **45**.
(28) Owing to extending and retracting of lift cylinders **44** and **45**, the rear end portion of draw bar **40** can move up and down with respect to front frame **22**. Draw bar **40** is vertically swingable with an axis along a direction of travel of the vehicle being defined as the center, as a result of extending and retracting of lift cylinders **44** and **45**. As a result of extending and retracting of a draw bar shift cylinder **46**, draw bar **40** is movable laterally with respect to front frame **22**.
(29) Lift cylinders **44** and **45** are attached to draw bar **40** and a bracket **50**. Lift cylinders **44** and **45** each have a head attached to bracket **50**. A tip end portion of a rod of each of lift cylinders **44** and **45** is attached to draw bar **40**. Bracket **50** is attached to front frame **22**.
(30) Draw bar shift cylinder **46** is attached to draw bar **40** and bracket **50**. A tip end portion of a head of draw bar shift cylinder **46** is attached to draw bar **40**. A tip end portion of a rod of draw bar shift cylinder **46** is attached to bracket **50**.
(31) Swing circle **41** is arranged below front frame **22**. Swing circle **41** is arranged below draw bar **40**. Swing circle **41** is revolvably (rotatably) supported on the rear end portion of draw bar **40**. Swing circle **41** can be driven by a slewing motor **49** as being revolvable clockwise and counterclockwise with respect to draw bar **40** when viewed from above the vehicle. Blade **42** is disposed on swing circle **41**. As swing circle **41** is driven to revolve, a blade propulsive angle of

blade **42** is adjusted. As will be described in detail later with reference to FIG. **4**, the blade propulsive angle refers to an angle of inclination of blade **42** with respect to the fore/aft direction of motor grader **1**.
(32) Blade **42** is arranged between front wheel **11** and rear wheel **12**. Front wheel **11** is arranged in front of blade **42**. Rear wheel **12** is arranged in the rear of blade **42**. Blade **42** is arranged between the front end of vehicular body frame **2** and a rear end of vehicular body frame **2**. Blade **42** is supported on swing circle **41**. Blade **42** is supported on draw bar **40** with swing circle **41** being interposed. Blade **42** is supported on front frame **22** with swing circle **41** and draw bar **40** being interposed.
(33) Blade **42** is supported as being movable in the lateral direction with respect to swing circle **41**. Specifically, a blade shift cylinder **47** is attached to swing circle **41** and blade **42** and arranged along a longitudinal direction of blade **42**. With blade shift cylinder **47**, blade **42** is movable in the lateral direction with respect to swing circle **41**. Blade **42** is movable in a direction intersecting with a longitudinal direction of front frame **22**.
(34) Blade **42** is supported as being swingable around an axis extending in the longitudinal direction of blade **42** with respect to swing circle **41**. Specifically, a tilt cylinder **48** is attached to swing circle **41** and blade **42**. As a result of extending and retracting of tilt cylinder **48**, blade **42** swings around the axis extending in the longitudinal direction of blade **42** with respect to swing circle **41**, so that an angle of inclination of blade **42** with respect to the direction of travel of the vehicle can be changed.
(35) As set forth above, blade **42** is constructed to be able to move up and down with respect to the vehicle, swing around the axis along the direction of travel of the vehicle, change an angle of inclination with respect to the fore/aft direction, move in the lateral direction, and swing around the axis extending in the longitudinal direction of blade **42**, with draw bar **40** and swing circle **41** being interposed.
(36) In the present example, a case in which the position of blade **42** with respect to swing circle **41** coincides with the neutral position and the position of draw bar **40** with respect to front frame **22** coincides with the neutral position is referred to as a “neutral position of work implement **4**.”
(37) <C. Functional Configuration>
(38) FIG. **3** is a functional block diagram illustrating a functional configuration of a control system of motor grader **1**.
(39) FIG. **3** shows relation between a main controller **150** and other peripheral devices. A work implement lever **118**, a switch **120**, a monitor apparatus **121**, a control valve **134**, sensors **171** and **174** to **177**, slewing motor **49**, lift cylinders **44** and **45**, and swing circle **41** are shown as the peripheral devices.
(40) Work implement **118**, switch **120**, and monitor apparatus **121** are provided in cab **3**.
(41) Main controller **150** is a controller that controls the entire motor grader **1**. Main controller **150** is implemented by a central processing unit (CPU), a non-volatile memory where a program is stored, and the like.
(42) Main controller **150** controls monitor apparatus **121**, control valve **134**, and the like.
(43) Monitor apparatus **121**, work implement lever **118**, and switch **120** are connected to main controller **150**.
(44) Main controller **150** provides a lever operation signal (an electrical signal) in accordance with an operated state of work implement lever **118** to control valve **134**.
(45) Control valve **134** is an electromagnetic proportional valve. Control valve **134** is connected to main controller **150**. Main controller **150** provides an operation signal (electrical signal) in accordance with a direction of operation and/or an amount of operation onto work implement lever **118** to control valve **134**. Control valve **134** controls an amount of hydraulic oil to be supplied from a hydraulic pump (not shown) to a hydraulic actuator in accordance with the operation signal. Exemplary hydraulic actuators include slewing motor **49**, lift cylinders **44** and **45**, draw bar shift

cylinder **46**, blade shift cylinder **47**, and tilt cylinder **48**.
(46) Main controller **150** includes a notification unit **153**, a memory **155**, and a control valve control unit **156**.
(47) Sensor **171** detects an angle of rotation (typically, a blade propulsive angle θ which will be described later) of swing circle **41**. Sensor **171** transmits information on the angle of rotation to control valve control unit **156**.
(48) Sensor **174** detects a cylinder length of lift cylinder **44**. Sensor **175** detects a cylinder length of lift cylinder **45**. Sensor **176** detects a cylinder length of draw bar shift cylinder **46**. Sensor **177** detects a cylinder length of blade shift cylinder **47**. Results detected by sensors **174** to **177** are transmitted to control valve control unit **156**.
(49) Notification unit **153** instructs monitor apparatus **121** to give guidance information in accordance with an instruction from control valve control unit **156**.
(50) Various types of information on engine output torque are stored in memory **155**. Information on an engine output torque curve is stored in memory **155**. A reference value of the cylinder length of draw bar shift cylinder **46** and a reference value of a cylinder length of blade shift cylinder **47** are stored in memory **155**.
(51) Control valve control unit **156** controls drive of slewing motor **49** by controlling opening of control valve **134** in accordance with magnitude of a current value which is an operation command to be provided. Control valve control unit **156** receives information on a circle rotation angle from sensor **171**. Control valve control unit **156** corrects a current value which is an operation command to control valve **134** based on the information on the circle rotation angle from sensor **171**.
(52) Switch **120** is a switch for automatic transition of work implement **4** to the neutral position. Switch **120** is a switch for automatic transition of the position of blade **42** with respect to swing circle **41** to the neutral position of blade **42** (which is also referred to as a "neutral position NB" below) with respect to swing circle **41**. Switch **120** is a switch for automatic transition of the position of draw bar **40** with respect to front frame **22** to the neutral position of draw bar **40** (which is also referred to as a "neutral position ND" below) with respect to front frame **22**. For example, a push button switch can be employed as switch **120**. Motor grader **1** may include a control lever instead of switch **120** for automatic transition of work implement **4** to the neutral position. Motor grader **1** should only include an operation apparatus for automatic transition of work implement **4** to the neutral position.
(53) <D. As to Neutral Position>
(54) FIG. **4** is a schematic diagram showing a state in which each of blade **42** and draw bar **40** is located at the neutral position.
(55) As shown in FIG. **4**, draw bar **40** moves in a direction shown with an arrow **903**. Swing circle **41** rotates in a direction shown with an arrow **902**. Blade **42** moves in a direction shown with an arrow **901**. Blade **42** rotates around a rotation axis C**1** by swing circle **41** being driven to revolve. As blade **42** rotates around rotation axis C**1**, a blade propulsive angle θ is varied. Blade propulsive angle θ refers to an angle formed between a direction of travel of the vehicular body and blade **42**. Blade propulsive angle θ refers to an angle of inclination of blade **42** with respect to the longitudinal direction of front frame **22**.
(56) In the following, for the sake of convenience of description, a virtual line orthogonal to rotation axis C**1** and in parallel to blade **42** (a centerline K of blade **42**) is defined as a line M**1**. A virtual line orthogonal to rotation axis C**1** and orthogonal to line M**1** is defined as a line M**2**. Line M**1** and line M**2** are assumed as lines in parallel to an XY plane.
(57) Initially, neutral position NB of blade **42** will be described.
(58) The position of blade **42** with respect to swing circle **41** when a central point C**2** in the longitudinal direction of blade **42** is located on line M**2** is neutral position NB. Central point C**2** is intermediate between a right end **421** and a left end **422** of blade **42**.
(59) Regardless of an angle of rotation of swing circle **41**, when central point C**2** is located on line

M**2**, the position of blade **42** with respect to swing circle **41** is neutral position NB. Regardless of a value of blade propulsive angle θ, when central point C**2** is located on line M**2**, the position of blade **42** with respect to swing circle **41** is neutral position NB.
(60) Regardless of the position of draw bar **40**, when central point C**2** is located on line M**2**, the position of blade **42** with respect to swing circle **41** is neutral position NB. Regardless of an attitude of draw bar **40**, when central point C**2** is located on line M**2**, the position of blade **42** with respect to swing circle **41** is neutral position NB.
(61) Neutral position ND of draw bar **40** will now be described.
(62) When lift cylinders **44** and **45** are identical to each other in cylinder length and rotation axis C**1** of swing circle **41** is located on an axial line J of front frame **22**, the position of draw bar **40** with respect to front frame **22** is neutral position ND.
(63) When lift cylinders **44** and **45** are identical to each other in cylinder length and axial line J of front frame **22** intersects with rotation axis C**1** of swing circle **41**, the position of draw bar **40** with respect to front frame **22** is neutral position ND.
(64) Regardless of an angle of rotation of swing circle **41**, draw bar **40** can be located at neutral position ND. Regardless of the position of blade **42** with respect to swing circle **41**, draw bar **40** can be located at neutral position ND.
(65) <E. Automatic Transition to Neutral Position>
(66) An operation of the motor grader when the operator operates switch **120** will be described. Specifically, an operation for automatic transition of work implement **4** to the neutral position will be described. By way of example, an operation to cause blade **42** to make transition to neutral position NB and thereafter to cause draw bar **40** to make transition to neutral position ND will be described.
(67) Specifically, when the operator operates switch **120**, an operation to cause blade **42** to make transition to neutral position NB and an operation to cause draw bar **40** to make transition to neutral position ND are successively performed in this order. Examples of the operation onto switch **120** include a press-and-hold operation (a press-down operation lasting for a certain time period or longer).
(68) Typically, when the operator operates switch **120**, on condition that motor grader **1** is traveling forward, main controller **150** causes work implement **4** to make automatic transition to the neutral position. During works, motor grader **1** is traveling forward. Therefore, when work implement **4** makes automatic transition to the neutral position while at least motor grader **1** is traveling forward, convenience of the operator is not compromised. By setting forward travel as the condition, work implement **4** does not make automatic transition to the neutral position even when switch **120** is operated in a standstill state.
(69) The condition for automatic transition of work implement **4** to the neutral position, however, does not necessarily require forward travel. Motor grader **1** may be configured such that work implement **4** makes automatic transition to the neutral position even when it is in a standstill state or is traveling rearward.
(70) Automatic transition to the neutral position is made by control by control valve control unit **156** within main controller **150**. Typically, automatic transition to the neutral position is made by execution of a program (control program) within the memory by the CPU. Automatic transition to the neutral position may be made by a semiconductor integrated circuit (an application specific integrated circuit (ASIC)).
(71) FIGS. **5** to **7** below show only a main part of motor grader **1** for facilitating understanding of operations of draw bar **40**, swing circle **41**, blade **42**, lift cylinders **44** and **45**, and draw bar shift cylinder **46**. FIGS. **5** to **7** do not show, for example, front wheel **11**.
e1. Transition of Blade **42** to Neutral Position NB
(72) FIG. **5** is a diagram for illustrating an operation for causing blade **42** to make transition to neutral position NB.

(73) Referring to FIG. **5**, in a state (A), the position of blade **42** with respect to swing circle **41** is displaced from neutral position NB to the right (the negative direction along the Y axis). In the state (A), the position of draw bar **40** with respect to front frame **22** is also displaced from neutral position ND to the right. Lift cylinder **44** is longer in cylinder length than lift cylinder **45**.
(74) When the operator operates switch **120** in the state (A), main controller **150** (specifically, control valve control unit **156**) causes blade shift cylinder **47** (see FIG. **2**) to operate such that the position of blade **42** with respect to swing circle **41** comes closer to neutral position NB. Main controller **150** causes blade shift cylinder **47** to operate such that the position of blade **42** reaches neutral position NB. When the position of blade **42** reaches neutral position NB, main controller **150** stops the operation of blade shift cylinder **47**.
(75) Through processing above, the state of motor grader **1** makes transition from the state (A) to a state (B). During transition from the state (A) to the state (B), the cylinder length of each of lift cylinders **44** and **45** and the cylinder length of draw bar shift cylinder **46** do not change. Main controller **150** causes blade shift cylinder **47** to operate such that the position of blade **42** comes closer to neutral position NB while the cylinder length of each of lift cylinders **44** and **45** is maintained at a length at the time when switch **120** is operated.
(76) More specific description is as below. In memory **155** (see FIG. **3**) of main controller **150**, the cylinder length (corresponding to the reference value above) of blade shift cylinder **47** at which the position of blade **42** coincides with neutral position NB is stored in advance. Main controller **150** causes blade shift cylinder **47** to operate based on the operation onto switch **120** until the cylinder length detected by sensor **177** (see FIG. **3**) attains to the cylinder length at which the position of blade **42** coincides with neutral position NB.
e2. Transition of Draw Bar **40** to Neutral Position ND
(77) Main controller **150** causes lift cylinders **44** and **45** and draw bar shift cylinder **46** to operate based on completion of the operation of blade shift cylinder **47** such that the position of draw bar **40** with respect to front frame **22** comes closer to neutral position ND. Main controller **150** causes lift cylinders **44** and **45** and draw bar shift cylinder **46** to operate such that the position of draw bar **40** with respect to front frame **22** reaches neutral position ND.
(78) FIG. **6** is a diagram for illustrating an operation for causing draw bar **40** to make transition to neutral position ND.
(79) Referring to FIG. **6**, a state (A) shows a state the same as the state (B) in FIG. **5**. Transition from the state (A) to a state (B) is based on the operation of lift cylinders **44** and **45**. Transition from the state (B) to a state (C) is based on the operation of draw bar shift cylinder **46**.
(80) (1) Operation of Lift Cylinders **44** and **45**
(81) Main controller **150** causes lift cylinders **44** and **45** to operate based on completion of the operation of blade shift cylinder **47** (the state (A)). Main controller **150** causes lift cylinders **44** and **45** to operate to thereby control the cylinder lengths of lift cylinders **44** and **45** to be equal to each other. Main controller **150** causes at least one of lift cylinders **44** and **45** to operate such that the cylinder length of lift cylinder **44** is equal to the cylinder length of lift cylinder **45**. Main controller **150** stops the operation of lift cylinders **44** and **45** when lift cylinder **44** and lift cylinder **45** become equal in cylinder length to each other as shown in the state (B).
(82) In the example in FIG. **6**, main controller **150** causes lift cylinder **44** to operate such that the cylinder length of lift cylinder **44** is equal to the cylinder length of lift cylinder **45**. Since lift cylinder **44** is longer in cylinder length than lift cylinder **45**, main controller **150** sets the cylinder length of lift cylinder **44** and the cylinder length of lift cylinder **45** to be equal to each other by decreasing the cylinder length of lift cylinder **44**. By thus controlling the cylinder length of one lift cylinder to be shorter, blade **42** can be prevented from digging into the ground at the time when draw bar shift cylinder **46** is caused to operate.
(83) Without being limited as such, main controller **150** may cause lift cylinder **45** to operate such that the cylinder length of lift cylinder **45** is equal to the cylinder length of lift cylinder **44**.

Alternatively, main controller **150** may cause lift cylinders **44** and **45** to operate such that the cylinder length of lift cylinder **44** and the cylinder length of lift cylinder **45** are set to a value (for example, an average value) between current cylinder lengths. According to such processing, a time period required for setting lift cylinder **45** and lift cylinder **44** to be equal in cylinder length to each other can be reduced.
(84) Through processing above, the state of motor grader **1** makes transition from the state (A) to the state (B). During transition from the state (A) to the state (B), the cylinder length of draw bar shift cylinder **46** and the cylinder length of blade shift cylinder **47** do not change.
(85) More specific description is as below. Main controller **150** determines a target cylinder length based on the cylinder length detected by sensor **174** (see FIG. **3**) and the cylinder length detected by sensor **175** at the time when switch **120** is operated. For example, main controller **150** determines the cylinder length detected by sensor **174** (or sensor **175**) as the target cylinder length. Alternatively, main controller **150** determines an average value of the cylinder length detected by sensor **174** and the cylinder length detected by sensor **175** as the target cylinder length.
(86) Main controller **150** controls the cylinder lengths of lift cylinders **44** and **45** to be equal to each other by controlling the cylinder length of each of lift cylinders **44** and **45** to the target cylinder length. When the cylinder length of one lift cylinder is set as the target cylinder length as above, only the other lift cylinder should only be caused to operate.
(87) (2) Operation of Draw Bar Shift Cylinder **46**
(88) Main controller **150** causes draw bar shift cylinder **46** to operate such that the position of draw bar **40** comes closer to neutral position ND on condition that the cylinder lengths of lift cylinders **44** and **45** have become equal to each other (the state (B)). Main controller **150** causes draw bar shift cylinder **46** to operate such that the position of draw bar **40** reaches neutral position ND. Main controller **150** stops the operation of draw bar shift cylinder **46** when the position of draw bar **40** reaches neutral position ND.
(89) Through processing above, the state of motor grader **1** makes transition from the state (B) to the state (C). During transition from the state (B) to the state (C), the cylinder lengths of lift cylinders **44** and **45** and the cylinder length of blade shift cylinder **47** do not change.
(90) More specific description is as below. The cylinder length (corresponding to the reference value above) of draw bar shift cylinder **46** at which the position of draw bar **40** coincides with neutral position ND at the time when the cylinder lengths of lift cylinders **44** and **45** are set to the target cylinder length is stored in advance in memory **155** of main controller **150**.
(91) Typically, the cylinder length of draw bar shift cylinder **46** at which the position of draw bar **40** coincides with neutral position ND within a numerical range of cylinder lengths where lift cylinders **44** and **45** are identical to each other in cylinder length is stored in advance in memory **155**. Specifically, the value within the numerical range and the cylinder length of draw bar shift cylinder **46** at which the position of draw bar **40** coincides with neutral position ND at the time when the former value is attained to are stored in association with each other. For example, the value within the numerical range and the cylinder length of draw bar shift cylinder **46** at which the position of draw bar **40** coincides with neutral position ND at the time when that value is attained to are stored in association with each other as a function or a data table in memory **155**.
(92) Thus, with main controller **150**, when the cylinder lengths (the identical cylinder lengths) of lift cylinders **44** and **45** are determined, the cylinder length of draw bar shift cylinder **46** at which the position of draw bar **40** coincides with neutral position ND is uniquely determined.
(93) Main controller **150** causes draw bar shift cylinder **46** to operate until the cylinder length detected by sensor **176** (see FIG. **3**) attains to the cylinder length at which the position of draw bar **40** coincides with neutral position ND.
(94) When the position of blade **42** reaches neutral position NB and the position of draw bar **40** with respect to front frame **22** reaches neutral position ND, main controller **150** may cause monitor apparatus **121** to show that work implement **4** has reached the neutral position. According to such

representation, the operator can know that work implement **4** has reached the neutral position.

e3. Advantage

(95) FIG. **7** is a diagram for illustrating an advantage obtained by causing work implement **4** to make transition to the neutral position during works.

(96) Referring to FIG. **7**, a state (A) in the center shows a state the same as the state (C) in FIG. **6**. A line P represents a position of a lower end of blade **42** in the state (A).

(97) A state (B) shows a state in which the cylinder length of lift cylinder **45** has been reduced by a prescribed length (any length) by an operation by the operator from the state (A). A state (C) shows a state in which the cylinder length of lift cylinder **44** has been reduced by the prescribed length from the state (A).

(98) The cylinder length of lift cylinder **45** in the state (B) is equal to the cylinder length of lift cylinder **44** in the state (C). The cylinder length of lift cylinder **44** in the state (B) and the cylinder length of lift cylinder **45** in the state (C) are equal to each other, because they do not change.

(99) In this case, an amount of upward movement (an amount of movement in the upward direction) of right end **421** of blade **42** in the state (B) is the same as an amount of upward movement of left end **422** of blade **42** in the state (C). An amount of lowering (an amount of movement in the downward direction) of left end **422** of blade **42** in the state (B) is the same as an amount of lowering of right end **421** of blade **42** in the state (C).

(100) Thus, when blade **42** is located at neutral position NB and draw bar **40** is located at neutral position ND as in the state (A), the amount of upward/downward movement of left end **422** of the blade at the time when the operator operates lift cylinder **44** by a certain amount is identical to the amount of upward/downward movement of right end **421** of the blade at the time when the operator operates lift cylinder **45** by an amount the same as the amount of movement of lift cylinder **44**.

(101) Therefore, movement of blade **42** to a desired position even by an operator who is not skilled in operation is facilitated as compared with a case in which blade **42** and draw bar **40** are not at neutral positions NB and ND, respectively. Therefore, motor grader **1** can lessen burdens imposed on the operator at the time when the operator operates work implement **4**.

(102) An advantage in the case in which blade **42** and draw bar **40** are set to neutral positions NB and ND, respectively, is described above. Even when only blade **42** of blade **42** and draw bar **40** is caused to return to neutral position NB, blade **42** is more easily moved to a desired position than when blade **42** and draw bar **40** are not located at neutral positions NB and ND, respectively. Even when only draw bar **40** of blade **42** and draw bar **40** is caused to return to neutral position ND, blade **42** is more easily moved to a desired position than when blade **42** and draw bar **40** are not located at neutral positions NB and ND, respectively. Therefore, even according to such a configuration, motor grader **1** can lessen burdens imposed on the operator at the time when the operator operates work implement **4**.

e4. Summary

(103) Thus, the present disclosure includes not only a configuration in which blade **42** and draw bar **40** are set to neutral positions NB and ND, respectively, but also a configuration in which only blade **42** is set to neutral position NB and a configuration in which only draw bar **40** is set to neutral position ND. The configuration of motor grader **1** will be summarized below from a point of view of two latter configurations.

(104) (1) Motor grader **1** includes switch **120**, swing circle **41**, blade **42** supported on swing circle **41**, blade shift cylinder **47** arranged along the longitudinal direction of blade **42**, blade shift cylinder **47** moving blade **42** in the lateral direction with respect to swing circle **41**, and main controller **150** that causes blade shift cylinder **47** to operate.

(105) Main controller **150** receives an operation signal from switch **120**. Main controller **150** causes blade shift cylinder **47** to operate based on the received operation signal such that the position of blade **42** with respect to swing circle **41** comes closer to neutral position NB of blade **42** with respect to swing circle **41**. More specifically, main controller **150** causes blade shift cylinder

47 to operate based on the operation performed onto switch **120** such that the position of blade **42** reaches neutral position NB.
(106) (2) Motor grader **1** includes switch **120**, front frame **22**, draw bar **40** swingably attached to front frame **22**, draw bar shift cylinder **46** attached to draw bar **40**, draw bar shift cylinder **46** moving draw bar **40** in the lateral direction with respect to front frame **22**, lift cylinders **44** and **45** attached to draw bar **40**, lift cylinders **44** and **45** moving draw bar **40** in the direction toward front frame **22** and the direction away from front frame **22**, and main controller **150** that causes draw bar shift cylinder **46** and lift cylinders **44** and **45** to operate.
(107) Main controller **150** receives an operation signal from switch **120**. Main controller **150** causes draw bar shift cylinder **46** and lift cylinders **44** and **45** to operate based on the received operation signal such that the position of draw bar **40** with respect to front frame **22** comes closer to neutral position ND of draw bar **40** with respect to front frame **22**. More specifically, main controller **150** causes draw bar shift cylinder **46** and lift cylinders **44** and **45** to operate based on the operation performed onto switch **120** such that the position of draw bar **40** reaches neutral position ND of draw bar **40**.
e5. Flow of Processing
(108) FIG. **8** is a flowchart for illustrating a flow of processing performed in motor grader **1**.
(109) Referring to FIG. **8**, in step S**1**, switch **120** accepts an operation by the operator. Main controller **150** thus receives the operation signal from switch **120**.
(110) When main controller **150** receives the operation signal from switch **120**, in step S**2**, main controller **150** causes blade shift cylinder **47** to operate such that the position of blade **42** with respect to swing circle **41** comes closer to neutral position NB of blade **42** with respect to swing circle **41**.
(111) In step S**3**, main controller **150** determines whether or not the position of blade **42** has reached neutral position NB. Specifically, main controller **150** determines whether or not the position of blade **42** has reached neutral position NB based on a result of detection by sensor **177** (see FIG. **3**).
(112) When main controller **150** determines that the position of the blade has not reached the neutral position (NO in step S**3**), the process returns to step S**2**. When main controller **150** determines that the position of the blade has reached the neutral position (YES in step S**3**), in step S**4**, main controller **150** causes draw bar shift cylinder **46** and lift cylinders **44** and **45** to operate such that the position of draw bar **40** with respect to front frame **22** comes closer to neutral position ND of draw bar **40** with respect to front frame **22**.
(113) In step S**5**, main controller **150** determines whether or not the position of draw bar **40** has reached neutral position ND. Specifically, main controller **150** determines whether or not the position of draw bar **40** has reached neutral position ND based on a result of detection by sensor **176**.
(114) When main controller **150** determines that the position of the draw bar has not reached the neutral position (NO in step S**5**), the process returns to step S**4**. When main controller **150** determines that the position of the draw bar has reached the neutral position (YES in step S**5**), main controller **150** quits a series of processing.
(115) FIG. **9** is a flowchart for illustrating details of processing in step S**4** in FIG. **8**.
(116) Referring to FIG. **9**, in step S**41**, main controller **150** causes lift cylinders **44** and **45** to operate. In step S**42**, main controller **150** determines whether or not lift cylinders **44** and **45** are equal in cylinder length to each other. Specifically, main controller **150** determines whether or not the cylinder length of lift cylinder **44** and the cylinder length of lift cylinder **45** are equal to each other based on results of detection by sensors **174** and **175** (see FIG. **3**).
(117) When main controller **150** determines that the cylinder lengths are not equal to each other (NO in step S**42**), the process returns to step S**41**. When main controller **150** determines that the cylinder lengths are equal to each other (YES in step S**42**), main controller **150** causes draw bar

shift cylinder **46** to operate such that the position of draw bar **40** comes closer to neutral position ND.

(118) In step S**44**, main controller **150** determines whether or not the position of draw bar **40** has reached neutral position ND. Specifically, main controller **150** determines whether or not the position of draw bar **40** has reached neutral position ND based on the result of detection by sensor **176** (see FIG. **3**).

(119) When main controller **150** determines that the position of draw bar **40** has not reached neutral position ND (NO in step S**44**), the process returns to step S**43**. When main controller **150** determines that the position of draw bar **40** has reached neutral position ND (YES in step S**44**), main controller **150** quits a series of processing.

(120) In exemplary processing above, blade **42** is moved to neutral position NB and thereafter draw bar **40** is moved to neutral position ND. Without being limited as such, draw bar **40** may be moved to neutral position ND and thereafter blade **42** may be moved to neutral position NB. Alternatively, an operation to move blade **42** to neutral position NB and an operation to move draw bar **40** to neutral position ND may simultaneously be performed based on the operation performed onto switch **120**. By simultaneously moving blade **42** and draw bar **40**, a time period for causing work implement **4** to return to the neutral position (causing blade **42** to return to neutral position NB and causing draw bar **40** to return to neutral position ND) can be reduced.

(121) “Being simultaneous” encompasses not only timing of start of movement of blade **42** being identical to timing of start of movement of draw bar **40** but also a state in which blade **42** is being moved and draw bar **40** is being moved at certain timing.

(122) In exemplary processing above, in movement of draw bar **40** to neutral position ND, lift cylinders **44** and **45** are caused to operate and thereafter draw bar shift cylinder **46** is caused to operate. Without being limited as such, draw bar shift cylinder **46** may be caused to operate and thereafter lift cylinders **44** and **45** may be caused to operate. Causing lift cylinders **44** and **45** to operate and thereafter causing draw bar shift cylinder **46** to operate can be less in digging of the ground by blade **42** than causing draw bar shift cylinder **46** to operate and thereafter causing lift cylinders **44** and **45** to operate.

(123) Lift cylinders **44** and **45** may be caused to operate simultaneously with draw bar shift cylinder **46**. By simultaneously causing lift cylinders **44** and **45** and draw bar shift cylinder **46** to operate, a time period for draw bar **40** to return to neutral position ND can be shortened.

(124) “Being simultaneous” encompasses not only timing of start of operations of lift cylinders **44** and **45** being identical to timing of start of an operation of draw bar shift cylinder **46** but also a state in which operations of lift cylinders **44** and **45** and an operation of draw bar shift cylinder **46** are being performed at certain timing.

(125) The embodiment disclosed herein is illustrative and not restricted to the above contents alone. The scope of the present invention is defined by the terms of the claims and is intended to include any modifications within the scope and meaning equivalent to the terms of the claims.

REFERENCE SIGNS LIST

(126) **1** motor grader; **2** vehicular body frame; **3** cab; **4** work implement; **6** engine compartment; **11** front wheel; **12** rear wheel; **21** rear frame; **22** front frame; **25** exterior cover; **40** draw bar; **41** swing circle; **42** blade; **44**, **45** lift cylinder; **46** draw bar shift cylinder; **47** blade shift cylinder; **48** tilt cylinder; **49** slewing motor; **50** bracket; **51** counter weight; **111** travel control lever; **118** work implement lever; **120** switch; **121** monitor apparatus; **134** control valve; **145** potentiometer; **146** starter switch; **148** transmission controller; **149** transmission; **150** main controller; **153** notification unit; **155** memory; **156** control valve control unit; **171**, **174**, **175**, **176**, **177** sensor; **402** ball bearing portion; **421** right end; **422** left end; C**1** rotation axis; C**2** central point; J**2** axial line; K centerline; M**1**, P line; NB, ND neutral position

Claims

1. A motor grader comprising: an operation apparatus; a swing circle; a blade supported on the swing circle and rotatable around a rotation axis by the swing circle; a first actuator arranged along a longitudinal direction of the blade, the first actuator moving the blade in a lateral direction with respect to the swing circle; and a controller that causes the first actuator to operate, wherein the controller receives an operation signal from the operation apparatus, and causes the first actuator to operate based on the received operation signal to move the blade in a horizontal direction with respect to the swing circle such that a position of the blade with respect to the swing circle comes closer to a neutral position of the blade with respect to the swing circle where a central point in the longitudinal direction of the blade is located on a line perpendicular to the rotation axis and the longitudinal direction and intersecting with the rotation axis in a top view of the motor grader.
2. The motor grader according to claim 1, wherein when the controller receives the operation signal from the operation apparatus, the controller causes the first actuator to operate such that the position of the blade comes closer to the neutral position of the blade on condition that the motor grader is traveling forward.
3. The motor grader according to claim 1, further comprising: a front frame; a draw bar attached to the front frame to fluctuate, the draw bar supporting the swing circle; and a second actuator attached to the draw bar, the second actuator moving the draw bar in a direction toward the front frame and a direction away from the front frame, wherein the second actuator is a lift cylinder, and when the controller receives the operation signal from the operation apparatus, the controller causes the first actuator to operate such that the position of the blade comes closer to the neutral position of the blade while a cylinder length of the lift cylinder is maintained at a cylinder length at time of reception of the operation signal.
4. The motor grader according to claim 3, further comprising a third actuator attached to the draw bar, the third actuator moving the draw bar in the lateral direction with respect to the front frame, wherein the controller causes the second actuator and the third actuator to operate based on completion of an operation of the first actuator, such that a position of the draw bar with respect to the front frame comes closer to a neutral position of the draw bar with respect to the front frame.
5. The motor grader according to claim 1, wherein the controller causes the first actuator to operate based on the received operation signal such that the position of the blade reaches the neutral position of the blade.
6. The motor grader according to claim 5, wherein the first actuator is a blade shift cylinder, the motor grader further comprises: a sensor that detects a cylinder length of the blade shift cylinder; and a storage where the cylinder length of the blade shift cylinder at which the position of the blade coincides with the neutral position of the blade is stored in advance, and the controller causes the blade shift cylinder to operate based on the received operation signal until the cylinder length detected by the sensor attains to the cylinder length stored in advance in the storage.
7. The motor grader according to claim 1, further comprising: a front frame; a draw bar attached to the front frame to fluctuate, the draw bar supporting the swing circle; a second actuator attached to the draw bar, the second actuator moving the draw bar in a direction toward the front frame and a direction away from the front frame; and a third actuator attached to the front frame and the draw bar, the third actuator moving the draw bar in the lateral direction with respect to the front frame, wherein the controller causes the second actuator and the third actuator to operate based on the received operation signal such that a position of the draw bar with respect to the front frame comes closer to a neutral position of the draw bar with respect to the front frame.
8. The motor grader according to claim 4, wherein the controller causes the second actuator and the third actuator to operate based on the received operation signal such that the position of the draw bar reaches the neutral position of the draw bar.

9. The motor grader according to claim 4, wherein on condition that the position of the draw bar has reached the neutral position of the draw bar, the controller causes the first actuator to operate such that the position of the blade comes closer to the neutral position of the blade.
10. The motor grader according to claim 4, wherein on condition that the position of the blade reaches the neutral position of the blade, the controller causes the second actuator and the third actuator to operate such that the position of the draw bar comes closer to the neutral position of the draw bar.
11. The motor grader according to claim 4, wherein the controller causes the first actuator, the second actuator, and the third actuator to simultaneously operate based on the received operation signal such that the position of the blade comes closer to the neutral position of the blade and the position of the draw bar comes closer to the neutral position of the draw bar.
12. A method of controlling a motor grader, the motor grader including an operation apparatus, a swing circle, a blade supported on the swing circle and rotatable around a rotation axis by the swing circle, and a first actuator arranged along a longitudinal direction of the blade, the first actuator moving the blade in a lateral direction with respect to the swing circle, the method comprising: receiving an operation signal from the operation apparatus based on an operation performed onto the operation apparatus; and causing the first actuator to operate based on reception of the operation signal from the operation apparatus to move the blade in a horizontal direction with respect to the swing circle such that a position of the blade with respect to the swing circle comes closer to a neutral position of the blade with respect to the swing circle where a central point in the longitudinal direction of the blade is located on a line perpendicular to the rotation axis and the longitudinal direction and intersecting with the rotation axis in a top view of the motor grader.
13. The method of controlling a motor grader according to claim 12, wherein the causing the first actuator to operate includes causing the first actuator to operate such that the position of the blade comes closer to the neutral position of the blade on condition that the motor grader is traveling forward.
14. The method of controlling a motor grader according to claim 12, wherein the motor grader further includes a draw bar attached to a front frame to fluctuate, the draw bar supporting the swing circle, and a second actuator attached to the draw bar, the second actuator moving the draw bar in a direction toward the front frame and a direction away from the front frame, and in the causing the first actuator to operate, the first actuator is caused to operate such that the position of the blade comes closer to the neutral position of the blade while a length of the second actuator is maintained at a length at time when the operation apparatus is operated.
15. The method of controlling a motor grader according to claim 14, wherein the motor grader further includes a third actuator attached to the draw bar, the third actuator moving the draw bar in a lateral direction with respect to the front frame, and the method further comprises causing the second actuator and the third actuator to operate based on completion of an operation of the first actuator such that a position of the draw bar with respect to the front frame comes closer to a neutral position of the draw bar with respect to the front frame.
16. The method of controlling a motor grader according to claim 12, wherein the causing the first actuator to operate includes causing the first actuator to operate such that the position of the blade reaches the neutral position of the blade.
17. The method of controlling a motor grader according to claim 16, wherein the first actuator is a blade shift cylinder, the motor grader further includes a sensor that detects a cylinder length of the blade shift cylinder, and a storage in which a cylinder length of the blade shift cylinder at which the position of the blade coincides with the neutral position of the blade is stored in advance, and the causing the first actuator to operate includes causing the blade shift cylinder to operate until the cylinder length detected by the sensor attains to the cylinder length stored in advance in the storage.
18. The method of controlling a motor grader according to claim 12, wherein the motor grader further includes a draw bar attached to a front frame to fluctuate, the draw bar supporting the swing

circle, a second actuator attached to the draw bar, the second actuator moving the draw bar in a direction toward the front frame and a direction away from the front frame, and a third actuator attached to the front frame and the draw bar, the third actuator moving the draw bar in a lateral direction with respect to the front frame, and the method further comprises causing the second actuator and the third actuator to operate based on the received operation signal such that a position of the draw bar with respect to the front frame comes closer to a neutral position of the draw bar with respect to the front frame.

19. The method of controlling a motor grader according to claim 15, wherein the causing the second actuator and the third actuator to operate includes causing the second actuator and the third actuator to operate such that the position of the draw bar reaches the neutral position of the draw bar.